

Sheth L.U.J. & Sir M.V College Of Science

Subject :- Cyber & Information Security

Practical No 4

Aim :- Implement digital signature algorithms such as RSA-based signatures, and verify the integrity and authenticity of digitally signed messages.

Code :- CLI

```
from cryptography.hazmat.primitives.asymmetric import rsa,  
padding
```

```
from cryptography.hazmat.primitives import hashes
```

```
private_key = rsa.generate_private_key(  
    public_exponent=65537,  
    key_size=2048  
)
```

```
public_key = private_key.public_key()
```

```
message = input("Enter the message: ").encode()
```

```
signature = private_key.sign(  
    message,  
    padding.PSS(  
        mgf=padding.MGF1(hashes.SHA256()),  
        salt_length=padding.PSS.MAX_LENGTH  
    ),  
    hashes.SHA256()  
)
```

```
print("\nOriginal Message:", message.decode())  
print("Digital signature:", signature.hex())
```

```
try:  
    public_key.verify(  
        signature,  
        message,  
        padding.PSS(  
            mgf=padding.MGF1(hashes.SHA256()),  
            salt_length=padding.PSS.MAX_LENGTH  
        ),  
    )
```

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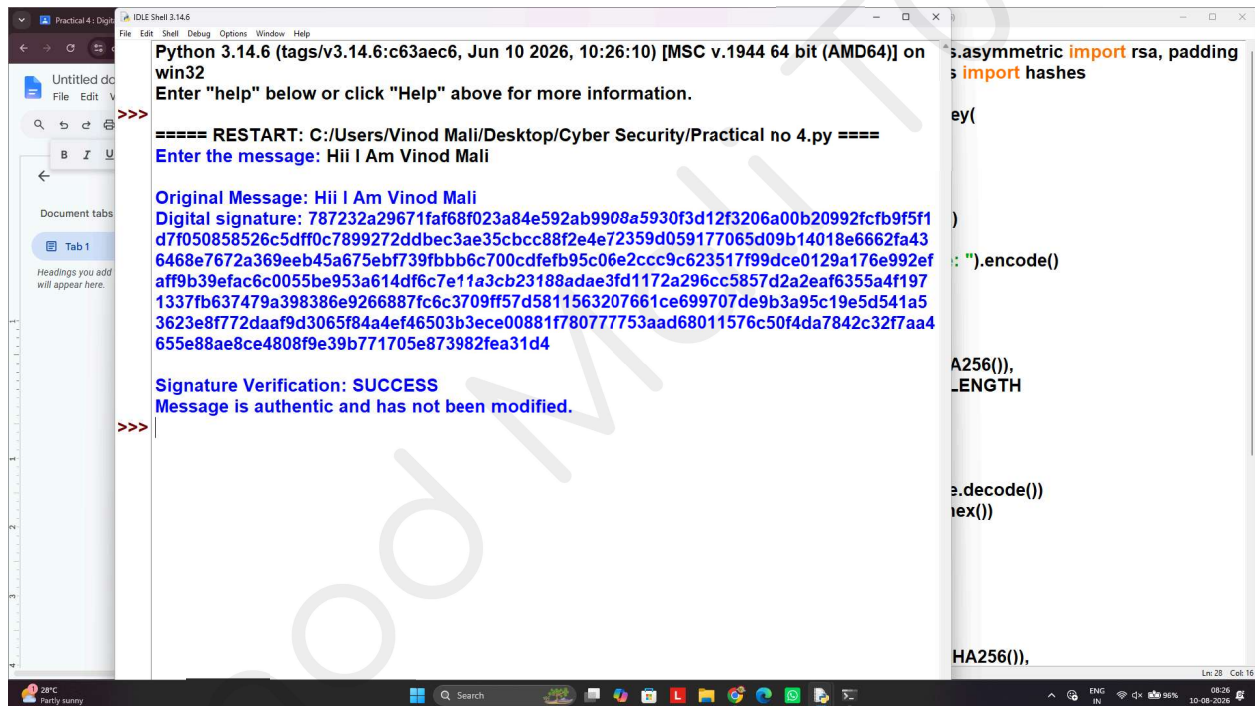
```
hashes.SHA256()  
)
```

```
print("\nSignature Verification: SUCCESS")  
print("Message is authentic and has not been modified.")
```

except Exception:

```
print("\nSignature Verification: FAILED")  
print("Message has been modified or signature is invalid.")
```

Output :-



```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32  
Enter "help" below or click "Help" above for more information.  
  
===== RESTART: C:/Users/Vinod Mali/Desktop/Cyber Security/Practical no 4.py =====  
Enter the message: Hii I Am Vinod Mali  
  
Original Message: Hii I Am Vinod Mali  
Digital signature: 787232a29671faf68f023a84e592ab9908a5930f3d12f3206a00b20992fcfb9f5f1  
d7f050858526c5dff0c7899272ddb3e3ae35cbcc88f2e4e72359d059177065d09b14018e6662fa43  
6468e7672a369eeb45a675ebf739fbbb6c700cdfefb95c06e2ccc9c623517f99dce0129a176e992ef  
aff9b39efac6c0055be953a614df6c7e11a3cb23188adae3fd1172a296cc5857d2a2eaf6355a4f197  
1337fb637479a398386e9266887fc6c3709ff57d5811563207661ce699707de9b3a95c19e5d541a5  
3623e8f772daaf9d3065f84a4ef46503b3ece00881f78077753aad68011576c50f4da7842c32f7aa4  
655e88ae8ce4808f9e39b771705e873982fea31d4  
  
Signature Verification: SUCCESS  
Message is authentic and has not been modified.  
>>>
```

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Code :- GUI

```
import tkinter as tk
from tkinter import messagebox
from cryptography.hazmat.primitives.asymmetric import rsa,
padding
from cryptography.hazmat.primitives import hashes

private_key = rsa.generate_private_key(
    public_exponent=65537,
    key_size=2048
)

public_key = private_key.public_key()

def generate_signature():
    message = message_entry.get()

    if message == "":
        messagebox.showwarning("Warning", "Please enter a message.")
        return

    global signature

    signature = private_key.sign(
        message.encode(),
        padding.PSS(
            mgf=padding.MGF1(hashes.SHA256()),
            salt_length=padding.PSS.MAX_LENGTH
        ),
        hashes.SHA256()
    )

    signature_text.delete("1.0", tk.END)
    signature_text.insert(tk.END, signature.hex())
```

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```
result_label.config(  
    text="Signature Generated Successfully",  
    fg="green"  
)
```

```
def verify_signature():  
    message = message_entry.get()  
  
    if message == "":  
        messagebox.showwarning("Warning", "Please enter a message.")  
        return  
  
    try:  
        public_key.verify(  
            signature,  
            message.encode(),  
            padding.PSS(  
                mgf=padding.MGF1(hashes.SHA256()),  
                salt_length=padding.PSS.MAX_LENGTH  
            ),  
            hashes.SHA256()  
        )  
  
        result_label.config(  
            text="✓ Signature Valid",  
            fg="green"  
        )  
  
    except:  
        result_label.config(  
            text="X Signature Invalid",  
            fg="red"  
        )
```

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```
def clear_fields():  
    message_entry.delete(0, tk.END)  
    signature_text.delete("1.0", tk.END)  
    result_label.config(text="")
```

```
root = tk.Tk()  
root.title("RSA Digital Signature")  
root.geometry("600x430")  
root.resizable(False, False)
```

```
tk.Label(  
    root,  
    text="RSA Digital Signature",  
    font=("Arial", 18, "bold")  
).pack(pady=15)
```

```
tk.Label(  
    root,  
    text="Message",  
    font=("Arial", 11)  
).pack()
```

```
message_entry = tk.Entry(  
    root,  
    width=60,  
    font=("Arial", 11)  
)  
message_entry.pack(pady=8)
```

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```
tk.Button(  
    root,  
    text="Generate Signature",  
    command=generate_signature,  
    width=20  
).pack(pady=8)
```

```
tk.Label(  
    root,  
    text="Digital Signature",  
    font=("Arial", 11)  
).pack()
```

```
signature_text = tk.Text(  
    root,  
    width=65,  
    height=6,  
    font=("Courier", 8)  
)  
signature_text.pack(pady=8)
```

```
button_frame = tk.Frame(root)  
button_frame.pack(pady=5)
```

```
tk.Button(  
    button_frame,  
    text="Verify",  
    command=verify_signature,  
    width=15  
).grid(row=0, column=0, padx=5)
```

```
tk.Button(  
    button_frame,  
    text="Clear",
```

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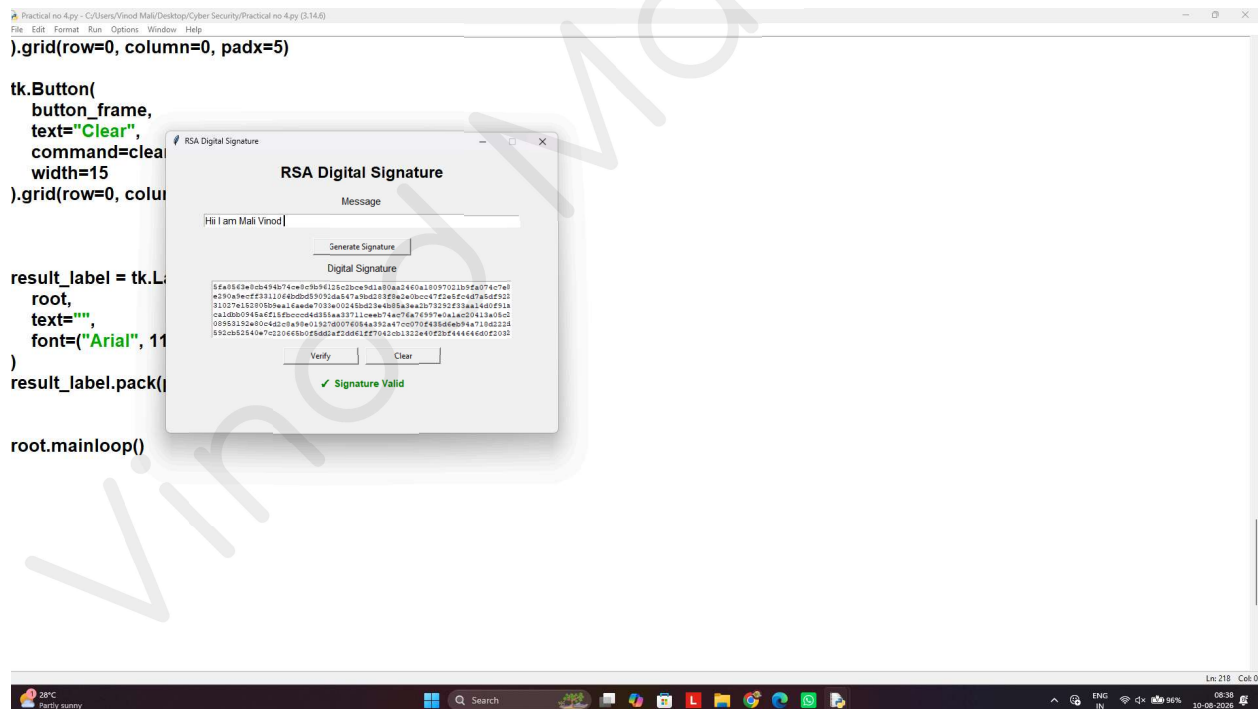
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```
command=clear_fields,  
width=15  
)grid(row=0, column=1, padx=5)
```

```
result_label = tk.Label(  
    root,  
    text="",  
    font=("Arial", 11, "bold")  
)  
result_label.pack(pady=12)
```

```
root.mainloop()
```

Output :-



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